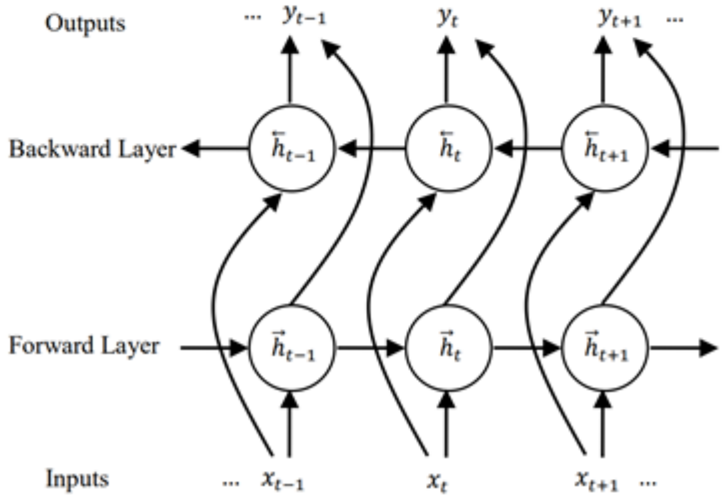
Back to Sentiment Analysis

- When using RNNs for sentence classification (such as sentiment analysis), it is often practical to use Bi-RNNs
- Bi-RNNs:
    - 2 RNNs, one going back to forth and the other forth to back
    - Output function is a function of both states
- This allows the states associated with each word to encode relevant information from the words following them and preceding them

$$
\begin{aligned}
\overrightarrow{h}_t &= R(W_{x\overrightarrow{h}} x_t + W_{\overrightarrow{h}\overrightarrow{h}} \overrightarrow{h}_{t-1} + b_{\overrightarrow{h}}) && (9)\\
\overleftarrow{h}_t &= R(W_{x\overleftarrow{h}} x_t + W_{\overleftarrow{h}\overleftarrow{h}} \overleftarrow{h}_{t+1} + b_{\overleftarrow{h}}) && (10)\\
y_t &= W_{\overrightarrow{h}y} \overrightarrow{h}_t + W_{\overleftarrow{h}y} \overleftarrow{h}_t + b_y && (11)
\end{aligned}
$$



"Contextual Bidirectional Long Short-Term Memory Recurrent Neural Network Language Models: A Generative Approach to Sentiment Analysis", Mousa and Shuller, EACL 2017

Back to Sentiment Analysis

- One simple way to do sentiment analysis (or other sentence classification) with Bi-RNNs is to average the output sequence:

$$y = \frac{1}{N} \sum_i y_i$$

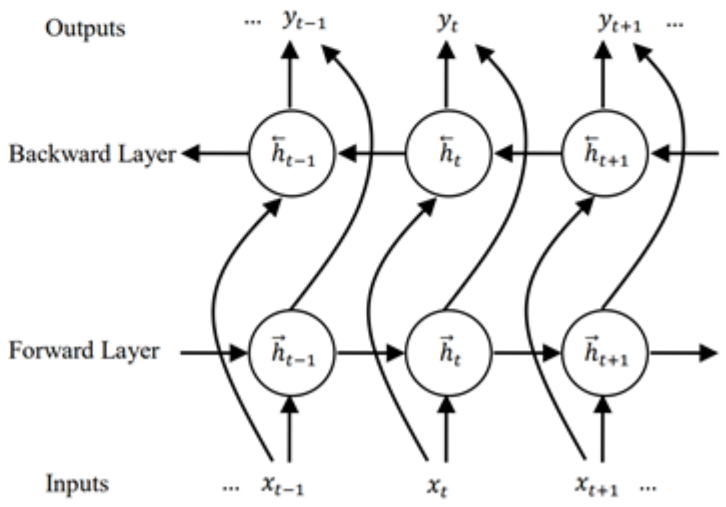
- Now train a binary (log-linear) classifier for predicting the sentiment:

$$P(+|y) = \frac{1}{1+e^{-(w^t \cdot y + b)}} = sigmoid(w^t \cdot y + b)$$

$$\overrightarrow{h}_t = R(W_{x\overrightarrow{h}} x_t + W_{\overrightarrow{h}\overrightarrow{h}} \overrightarrow{h}_{t-1} + b_{\overrightarrow{h}}) \quad (9)$$

$$\overleftarrow{h}_t = R(W_{x\overleftarrow{h}} x_t + W_{\overleftarrow{h}\overleftarrow{h}} \overleftarrow{h}_{t+1} + b_{\overleftarrow{h}}) \quad (10)$$

$$y_t = W_{\overrightarrow{h}y} \overrightarrow{h}_t + W_{\overleftarrow{h}y} \overleftarrow{h}_t + b_y \quad (11)$$



Context in RNN models for Sentiment Analysis

- RNN models are able to take the context (both preceding and following) into account, as well as the linear order between the words
  - bag-of-words models cannot
- They indeed show better performance in tasks such as sentiment analysis
- However, the state sequence is not easy to interpret
  - Much heavier computationally, especially training
- Further investigation is needed to establish what contextual and semantic aspects of sentences are captured using these techniques